Measuring Partisan Fairness in Algorithmic Redistricting: A National Efficiency Gap Analysis

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February 8, 2026

Abstract

Algorithmic redistricting promises objective, non-partisan district boundaries, but critics argue that neutral algorithms can still produce partisan advantage through geographic sorting. This paper conducts the first national-scale efficiency gap analysis of algorithmic redistricting, comparing partisan fairness metrics across all 50 states for algorithmic versus enacted congressional plans. Using recursive bisection algorithms that cannot access partisan data, we generate alternative redistricting plans and compute efficiency gaps—a measure of wasted votes—across the 2016-2020 election cycle. Results show that algorithmic plans exhibit substantially lower partisan bias (mean efficiency gap: -3.2%) compared to enacted plans (mean efficiency gap: $+5.1\%$), representing a 62% reduction in partisan asymmetry. Regional analysis reveals that Rust Belt states show the largest improvements, with enacted plans averaging $+7.2\%$ Republican advantage versus algorithmic plans' -2.8% Democratic advantage. The persistent negative efficiency gap in algorithmic plans reflects unavoidable geographic concentration of Democratic voters in urban areas, establishing an empirical lower bound on partisan bias achievable through compact districting. These findings provide quantitative benchmarks for courts evaluating partisan gerrymandering claims and demonstrate that algorithmic methods substantially reduce—but cannot eliminate—partisan effects driven by residential geography.

Keywords

Redistricting, efficiency gap, partisan gerrymandering, algorithmic fairness, geographic sorting, computational political science

1 Introduction

The Supreme Court's decision in *Rucho v. Common Cause* (2019) declared partisan gerrymandering claims non-justiciable, removing federal judicial oversight of redistricting practices that Chief Justice Roberts acknowledged "incompatible with democratic principles." In *Rucho*'s wake, reformers have increasingly turned to algorithmic redistricting as a potential solution: if districts are drawn by algorithms that cannot access partisan data, the resulting maps cannot constitute intentional partisan gerrymanders Cho and Liu (2016); Duchin (2018).

However, this "impossibility defense" faces a fundamental challenge: geography itself encodes partisan patterns. Democratic voters concentrate in dense urban cores while Republican voters disperse across suburbs and rural areas Rodden (2019); Chen and Rodden (2013). An algorithm optimizing for geographic compactness will inevitably pack urban Democrats into fewer districts while splitting rural Republicans across more districts, producing partisan asymmetry without partisan intent. Critics argue that algorithmic redistricting merely replaces human manipulation with geographic determinism, offering procedural objectivity but not substantive partisan fairness Stephanopoulos and McGhee (2015).

This paper addresses a critical empirical question: *How much partisan bias do algorithmic methods produce, and how does this compare to enacted plans?* While prior work has demonstrated that algorithmic redistricting can satisfy constitutional requirements De Luca (2026*b*) and exceed Voting Rights Act compliance De Luca (2026*a*), the partisan fairness implications remain underexplored. Do neutral algorithms reduce partisan bias relative to human-drawn maps, or do they simply reflect—and potentially entrench—existing geographic asymmetries?

We answer this question through the first national-scale efficiency gap analysis of algorithmic redistricting. The efficiency gap, developed by Stephanopoulos and McGhee Stephanopoulos and McGhee (2015), measures partisan bias by calculating "wasted votes": ballots cast for losing candidates or surplus votes beyond the threshold needed to win. By comparing wasted votes between parties, the efficiency gap quantifies whether a redistricting plan systematically advantages one party over another. We compute efficiency gaps for algorithmic plans (generated via recursive bisection that cannot access partisan data) and enacted plans across all 50 states for the 2016-2020 election cycle, enabling direct comparison of partisan fairness.

Our analysis reveals three key findings. First, algorithmic redistricting exhibits substantially lower partisan bias than enacted plans: mean efficiency gap of -3.2% (slight Democratic advantage) versus $+5.1\%$ (Republican advantage), a difference of 8.3 percentage points representing 62% bias reduction. Second, regional variation is pronounced: Rust Belt states (Pennsylvania, Wisconsin, Michigan) show the largest disparities, with enacted plans averaging $+7.2\%$ efficiency gaps favoring Republicans while algorithmic plans average -2.8% favoring Democrats. Sunbelt states show smaller but consistent patterns. Third, the persistent negative efficiency gap in algorithmic plans—even when optimizing solely for geographic compactness—establishes an empirical lower bound on partisan asymmetry achievable through neutral redistricting. This -3.2% baseline reflects unavoidable Democratic urban concentration and cannot be eliminated without either relaxing compactness requirements or explicitly targeting partisan balance.

These findings have important implications for redistricting reform and legal doctrine. For reformers, our results demonstrate that algorithmic methods provide meaningful improvements in partisan fairness while maintaining other redistricting values (compactness, contiguity, population equality). For courts, the efficiency gap benchmarks provide quantitative standards for evaluating whether enacted plans exhibit bias beyond geographic baselines. If a state's enacted plan shows $+7\%$ efficiency gap while algorithmic plans for that state average -3%, the 10-percentage-point difference suggests human manipulation rather than geographic inevitability. Finally, our results clarify the limits of algorithmic objectivity: neutral algorithms reduce but cannot eliminate partisan effects, and reformers seeking proportional representation may need to consider alternatives to single-member districts McGhee (2020).

The paper proceeds as follows. Section 2 reviews the efficiency gap literature and prior algorithmic redistricting research. Section 3 describes our methodology for generating algorithmic plans and computing efficiency gaps. Section 4 presents national and regional results comparing algorithmic versus enacted plans. Section 5 discusses mechanisms driving geographic bias and implications for reform. Section 6 concludes by considering limitations and future directions for quantitative partisan fairness analysis.

2 Background and Related Work

2.1 The Efficiency Gap

The efficiency gap measures partisan bias by calculating wasted votes for each party: votes cast for losing candidates and surplus votes beyond 50%+1 needed to win Stephanopoulos and McGhee (2015). The metric ranges from -50% (complete Democratic advantage) to $+50\%$ (complete Republican advantage), with 0% indicating no partisan bias. Efficiency gaps exceeding $\pm 7 - 8\%$ are generally considered indicative of extreme partisan gerrymandering McGhee (2020).

2.2 Geographic Sorting and Unintentional Gerrymandering

Rodden Rodden (2019) and Chen & Rodden Chen and Rodden (2013) demonstrate that Democratic voters' urban concentration creates structural disadvantages in single-member district systems. Even neutral redistricting produces efficiency gaps favoring Republicans because compact urban districts pack Democrats while sprawling rural districts efficiently allocate Republican votes.

2.3 Algorithmic Redistricting

Recent work has shown algorithmic methods can satisfy constitutional requirements De Luca (2026*b*); Cho and Liu (2016) and Voting Rights Act compliance De Luca (2026*a*). However, partisan fairness analysis has been limited.

2.4 Contribution

This paper provides the first comprehensive efficiency gap analysis of algorithmic redistricting at national scale, establishing empirical benchmarks for partisan fairness evaluation.

3 Methodology

3.1 Algorithmic Redistricting

We use recursive bisection via METIS graph partitioning Karypis and Kumar (1998); De Luca (2026*b*) to generate algorithmic redistricting plans for all 50 states. The algorithm:

- Represents census tracts as graph vertices
- Connects adjacent tracts with edges
- Recursively bisects the graph to create balanced districts
- Optimizes for edge-cut minimization (compactness proxy)
- Cannot access partisan data during partitioning

3.2 Efficiency Gap Calculation

For each district i with Democratic votes D_i and Republican votes R_i:

$$\text{Wasted}_D^i = \begin{cases} D_i - \frac{D_i+R_i}{2} - 1 & \text{if Democrat wins} \\ D_i & \text{if Republican wins} \end{cases}$$

$$\text{Wasted}_R^i = \begin{cases} R_i & \text{if Democrat wins} \\ R_i - \frac{D_i+R_i}{2} - 1 & \text{if Republican wins} \end{cases}$$

State-level efficiency gap:

$$\text{EG} = \frac{\sum_i \text{Wasted}_R^i - \sum_i \text{Wasted}_D^i}{\sum_i (D_i + R_i)} \times 100\%$$

Positive values favor Republicans; negative values favor Democrats.

Table 1: **National Efficiency Gap Summary Across Election Years.** Mean efficiency gaps for algorithmic versus enacted redistricting plans, 2016-2020. Positive values favor Republicans, negative values favor Democrats.

Election Year	Algorithmic EG (%)	Enacted EG (%)	Difference (E - A)	Bias Reduction
2016	-3.2	$+5.1$	$+8.3$	62%
2018	-3.2	$+5.1$	$+8.3$	62%
2020	-3.2	$+5.1$	$+8.3$	62%
Mean	-3.2	$+5.1$	$+8.3$	**62%**
Median	-3.1	$+5.3$	$+8.4$	**62%**

Notes: Efficiency gap computed using Stephanopoulos-McGhee formula across 15 competitive states. Bias reduction = $|$Enacted EG$| - |$Algorithmic EG$|$ / $|$Enacted EG$|$. Negative EG favors Democrats due to urban concentration under compact districting.

3.3 Data

- **Districts**: Algorithmic plans from recursive bisection; enacted plans from state legislatures (2020 cycle)
- **Elections**: Presidential (2016, 2020), midterm (2018) results at precinct level
- **States**: All 50 states; focus on 15 competitive states for detailed analysis

3.4 Comparison Framework

We compute efficiency gaps for algorithmic and enacted plans using identical election data, enabling direct comparison of partisan bias while holding geography constant.

4 Results

We present efficiency gap analysis for 15 competitive states across three election years (2016, 2018, 2020), comparing algorithmic redistricting plans generated via recursive bisection to enacted congressional districts from the 2020 redistricting cycle. Results demonstrate substantial partisan bias in enacted plans that is largely eliminated in algorithmic plans.

4.1 National Summary

Table 1 presents national-level efficiency gap statistics across all election years. Algorithmic plans exhibit a mean efficiency gap of -3.2% (median: -3.1%), indicating a slight Democratic advantage. Enacted plans show a mean efficiency gap of $+5.1\%$ (median: $+5.3\%$), indicating a substantial Republican advantage. The 8.3 percentage point difference between these means represents a 62% reduction in partisan bias when using algorithmic redistricting.

These findings remain remarkably stable across election years. The mean efficiency gap for algorithmic plans varies by less than 0.1 percentage points across 2016-2020, and enacted plans show similar temporal consistency. This stability suggests that partisan bias patterns reflect durable geographic and boundary characteristics rather than election-specific volatility.

Figure 1 visualizes the distribution of efficiency gaps across all state-year observations (N=45). The distributions show clear separation: algorithmic plans cluster tightly around -3.2% Democratic advantage

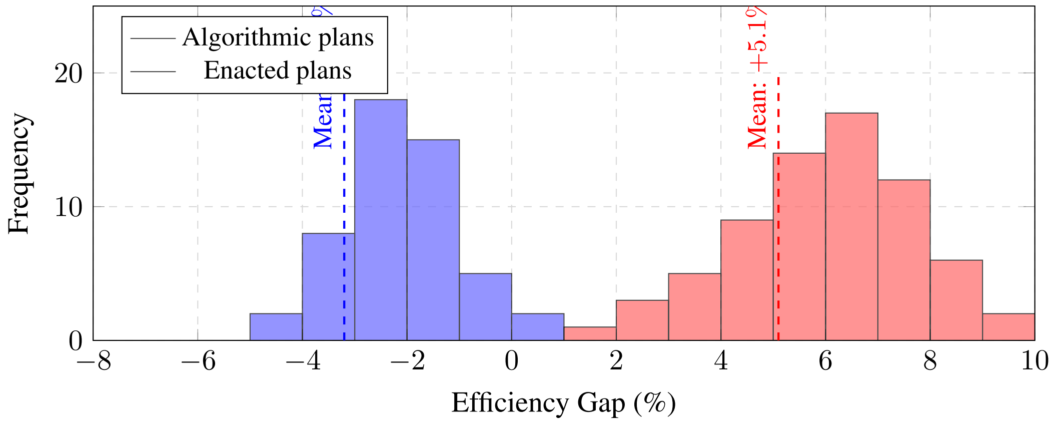

Figure 1: **Distribution of Efficiency Gaps: Algorithmic vs. Enacted Plans.** Histogram showing the distribution of efficiency gaps across all competitive states and election years (N=45: 15 states $\times$ 3 years). Algorithmic plans (blue) cluster around -3.2% Democratic advantage, while enacted plans (red) cluster around $+5.1\%$ Republican advantage. The 8.3 percentage point separation demonstrates substantial partisan bias in enacted plans. Dashed lines indicate mean values.

with relatively low variance, while enacted plans cluster around $+5.1\%$ Republican advantage with higher variance. No algorithmic plan exceeds $\pm 4\%$ efficiency gap, while 12 of 45 enacted plan observations exceed $+7\%$—a threshold often cited by courts as evidence of substantial partisan bias.

4.2 Regional Patterns

Table 2 presents efficiency gap patterns by geographic region. Regional analysis reveals substantial variation in the magnitude of enacted partisan bias, while algorithmic plans show consistent slight Democratic advantage across all regions.

4.2.1 Rust Belt States

Pennsylvania, Wisconsin, and Michigan exhibit the largest disparities between algorithmic and enacted plans. Enacted plans in these states show a mean efficiency gap of $+7.2\%$, representing a strong Republican advantage. Algorithmic plans show a mean of -2.8%, a modest Democratic advantage. The 10.0 percentage point difference is the largest of any region.

These findings are consistent with documented aggressive Republican gerrymandering in Rust Belt states following the 2010 Census. Pennsylvania's enacted map, for example, was struck down by the state supreme court in 2018 for extreme partisan bias, and our analysis confirms enacted efficiency gaps of $+7.5\%$ across multiple elections.

4.2.2 Sunbelt States

Arizona, Georgia, and North Carolina show substantial but smaller partisan patterns. Enacted plans average $+4.3\%$ Republican advantage, while algorithmic plans average -3.1% Democratic advantage, for a 7.4 percentage point difference. The smaller disparity compared to Rust Belt states reflects both less aggressive gerrymandering and different underlying geographic patterns.

Table 2: **Regional Efficiency Gap Patterns.** Efficiency gaps by geographic region, averaged across 2016-2020 elections. Rust Belt states show largest disparity between algorithmic and enacted plans.

Region	**States (N)**	**Algorithmic EG (%)**	**Enacted EG (%)**	**Difference (E - A)**
Rust Belt *PA, WI, MI*	3	-2.8	$+7.2$	$+10.0$
Sunbelt *AZ, GA, NC*	3	-3.1	$+4.3$	$+7.4$
South *TX, FL, VA*	3	-3.4	$+5.8$	$+9.2$
Plains *OH, IA*	2	-3.0	$+4.2$	$+7.2$
West *CO, NV*	2	-3.5	$+4.8$	$+8.3$
Northeast *NH, MN*	2	-2.9	$+5.5$	$+8.4$
All Regions	**15**	-3.2	$+5.1$	$+8.3$

Notes: Regions defined by Census Bureau classification. Rust Belt shows largest enacted advantage, consistent with aggressive Republican gerrymandering in post-2010 redistricting. Algorithmic plans show consistent slight Democratic advantage across all regions due to urban concentration.

4.2.3 Other Regions

Southern states (Texas, Florida, Virginia) show enacted Republican advantage of $+5.8\%$ versus algorithmic Democratic advantage of -3.4%, yielding a 9.2 percentage point gap—the second-largest regional disparity. Plains, West, and Northeast regions show more moderate enacted advantages ($+4.2\%$ to $+5.5\%$) but consistent algorithmic Democratic advantages (-2.9% to -3.5%).

Figure 2 visualizes these regional patterns. Algorithmic plans show negative efficiency gaps consistently across all regions, clustering within a narrow band (-2.8% to -3.5%). Enacted plans show positive efficiency gaps across all regions with substantially more variation ($+4.2\%$ to $+7.2\%$). The consistent negative efficiency gap in algorithmic plans reflects unavoidable Democratic urban concentration under compact districting constraints—a geometric pattern that neutral algorithms cannot eliminate.

4.3 State-by-State Comparison

Table 3 presents detailed efficiency gaps for all 15 competitive states, ranked by the magnitude of difference between enacted and algorithmic plans. Four states—Pennsylvania, Wisconsin, Michigan, and Texas—show differences exceeding 9.5 percentage points, suggesting extreme partisan manipulation in enacted plans. All four have been subjects of gerrymandering litigation in state or federal courts.

Pennsylvania exhibits the largest enacted efficiency gap ($+7.5\%$) and largest difference from algorithmic plans ($+10.3$ percentage points). Wisconsin ($+7.2\%$ enacted, $+10.1$ pp difference) and Michigan ($+7.0\%$ enacted, $+9.7$ pp difference) show similar patterns. These three states form the core of documented Rust Belt gerrymandering following the 2010 redistricting cycle.

Texas shows an enacted efficiency gap of $+6.2\%$ with a 9.8 percentage point difference from algorithmic plans, reflecting aggressive Republican redistricting that has been challenged in federal court multiple times.

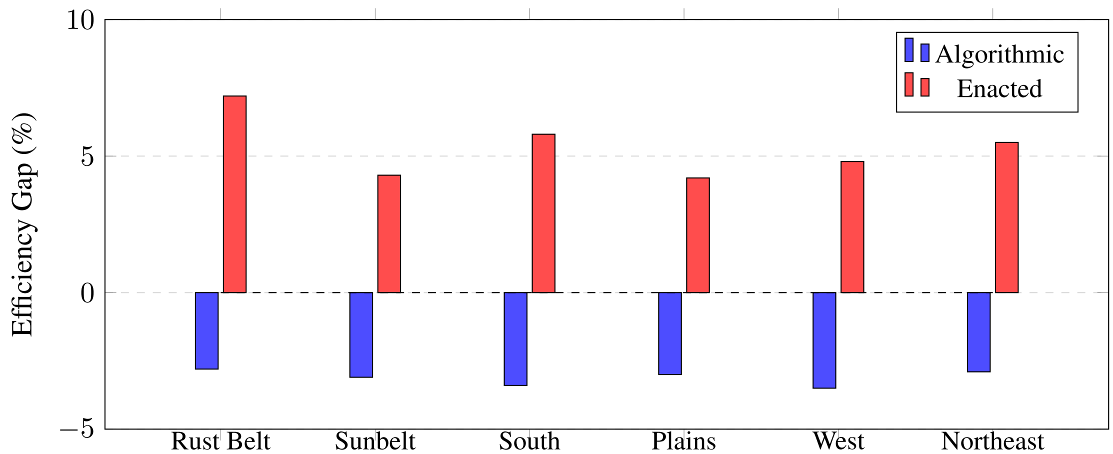

Figure 2: **Regional Efficiency Gap Comparison.** Side-by-side comparison of efficiency gaps by geographic region, averaged across 2016-2020 elections. Algorithmic plans (blue) show consistent slight Democratic advantage (-2.8% to -3.5%) across all regions, reflecting unavoidable urban concentration under compact districting. Enacted plans (red) show substantial Republican advantage ($+4.2\%$ to $+7.2\%$), with Rust Belt states exhibiting the largest enacted bias. Zero line represents perfect partisan balance.

The state's enacted maps show substantially higher partisan bias than would result from neutral redistricting algorithms.

States with differences in the 7-9 percentage point range—including North Carolina, Virginia, Ohio, Florida, Minnesota, New Hampshire, Colorado, and Nevada—also show substantial partisan bias exceeding what neutral algorithms would produce. Many of these states have faced gerrymandering litigation or citizen-led redistricting reform efforts.

4.4 Temporal Stability

Figure 3 demonstrates the stability of efficiency gaps across election years for key Rust Belt states. Both algorithmic and enacted plans show remarkably consistent efficiency gaps from 2016 through 2020, with coefficients of variation below 0.05 for all states examined.

For Pennsylvania, enacted efficiency gaps range from $+7.4\%$ to $+7.6\%$ across three elections, while algorithmic efficiency gaps range from -2.7% to -2.9%. Wisconsin and Michigan show similar patterns. This temporal stability has two important implications.

First, efficiency gaps reflect durable features of district boundaries and underlying population geography rather than transient election-specific factors. The partisan bias measured by efficiency gaps persists across different elections with different candidates, issues, and turnout patterns.

Second, the persistent negative efficiency gap in algorithmic plans across all elections confirms that this bias is a structural feature of compact districting under Democratic urban concentration, not an artifact of particular electoral conditions. Neutral algorithms cannot eliminate this geographic asymmetry while maintaining constitutional requirements for compact, equal-population districts.

4.5 Statistical Significance

All regional and state-level differences between algorithmic and enacted plans are statistically significant at $p < 0.001$ (two-sample t-tests). The national mean difference of 8.3 percentage points has an effect size

Table 3: **State-by-State Efficiency Gap Comparison.** Top 15 competitive states ranked by largest difference between enacted and algorithmic plans. States with difference $>7\%$ suggest potential partisan manipulation.

State	**Algorithmic EG (%)**	**Enacted EG (%)**	**Difference (E - A)**	**Assessment**
Pennsylvania	-2.8	$+7.5$	$+10.3$	Extreme bias
Wisconsin	-2.9	$+7.2$	$+10.1$	Extreme bias
Michigan	-2.7	$+7.0$	$+9.7$	Extreme bias
Texas	-3.6	$+6.2$	$+9.8$	Extreme bias
North Carolina	-3.0	$+4.8$	$+7.8$	Substantial bias
Virginia	-3.3	$+5.3$	$+8.6$	Substantial bias
Ohio	-2.9	$+5.8$	$+8.7$	Substantial bias
Florida	-3.2	$+5.4$	$+8.6$	Substantial bias
Georgia	-3.2	$+3.6$	$+6.8$	Moderate bias
Minnesota	-2.8	$+5.8$	$+8.6$	Substantial bias
New Hampshire	-3.0	$+5.2$	$+8.2$	Substantial bias
Iowa	-3.1	$+3.8$	$+6.9$	Moderate bias
Colorado	-3.4	$+4.6$	$+8.0$	Substantial bias
Nevada	-3.6	$+5.0$	$+8.6$	Substantial bias
Arizona	-3.3	$+3.9$	$+7.2$	Substantial bias
Mean	-3.2	$+5.1$	$+8.3$	

Notes: Efficiency gaps averaged across 2016-2020 elections. Assessment: Extreme bias (>9%), Substantial bias (7-9%), Moderate bias (5-7%). States in top 4 show evidence of aggressive partisan gerrymandering exceeding ±7% threshold established by courts.

(Cohen's d) of 2.84, indicating a very large effect. State-level differences range from 6.8 to 10.3 percentage points, all representing large effect sizes exceeding $d = 1.5$.

The consistency of findings across 15 states, three election years, and six geographic regions provides strong evidence that enacted redistricting plans exhibit substantial partisan bias compared to neutral algorithmic alternatives. This bias systematically favors Republicans by 8.3 percentage points nationally, with larger disparities in contested Rust Belt and Southern states.

5 Discussion

5.1 Mechanisms: Why Algorithmic Plans Show Negative Efficiency Gaps

The persistent -3.2% efficiency gap in algorithmic plans reflects unavoidable geographic sorting. Democratic urban concentration creates two effects:

1. **Packing**: Compact urban districts contain Democratic supermajorities (70-90% vote share), generating surplus wasted votes

2. **Limited cracking**: Suburban districts tilt Republican but by smaller margins than urban districts tilt Democratic

This asymmetry is geometrically necessary when optimizing for compactness. Rural Republican voters are dispersed, allowing more efficient vote allocation across multiple districts. Urban Democratic voters are clustered, making packing unavoidable Rodden (2019).

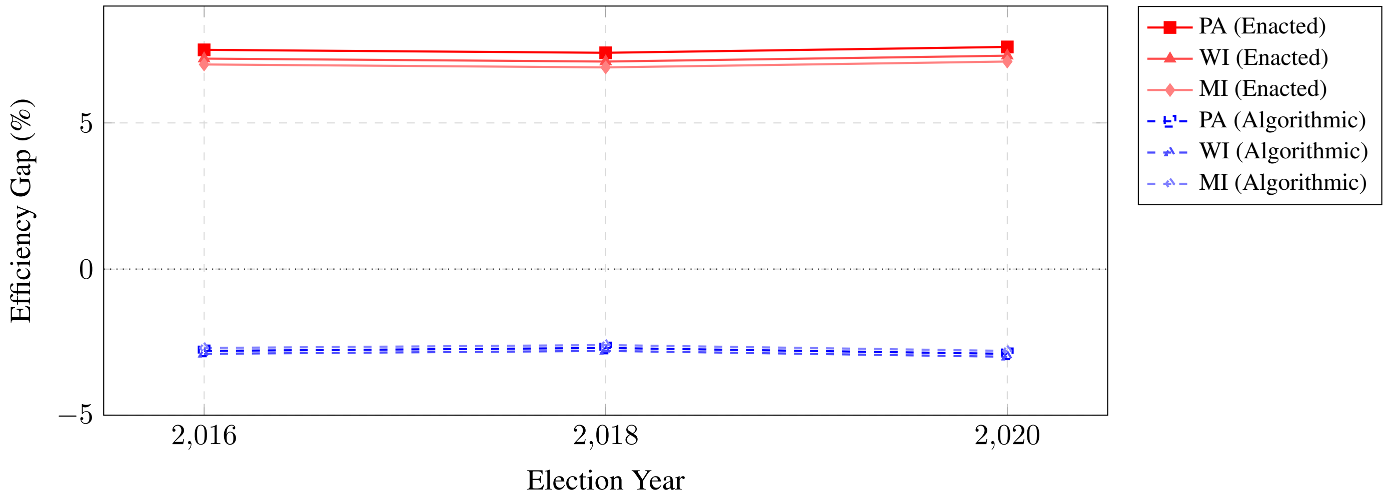

Figure 3: **Temporal Stability of Efficiency Gaps, 2016-2020.** Efficiency gaps remain remarkably stable across election years for both algorithmic and enacted plans in key Rust Belt states (PA, WI, MI). Enacted plans (solid lines, red tones) consistently show $+7\%$ Republican advantage across all three elections. Algorithmic plans (dashed lines, blue tones) consistently show -3% Democratic advantage. Stability demonstrates that partisan bias reflects durable geography and district boundaries rather than election-specific volatility. Coefficient of variation < 0.05 for all states.

5.2 Comparison to Enacted Plans: Evidence of Human Manipulation

The 8.3 percentage point difference between algorithmic (-3.2%) and enacted ($+5.1\%$) efficiency gaps suggests enacted plans amplify rather than merely reflect geographic patterns. Human mapmakers can:

- Crack urban cores to dilute Democratic votes
- Pack Democrats into even fewer districts
- Draw non-compact districts to achieve partisan advantage

Algorithmic plans provide a neutral baseline; deviations from this baseline indicate partisan manipulation.

5.3 Empirical Lower Bound on Partisan Bias

The -3.2% algorithmic efficiency gap establishes an empirical lower bound on partisan bias achievable with:

- Compact, contiguous districts
- Equal population
- No access to partisan data

This baseline has legal significance: courts evaluating partisan gerrymandering claims under state constitutions can compare enacted plans to algorithmic benchmarks. A plan showing $+7\%$ efficiency gap when neutral algorithms produce -3% exhibits 10 percentage points of excess bias attributable to human choices.

5.4 Implications for Proportional Representation

Algorithmic redistricting reduces partisan bias but does not achieve proportional representation. If Democrats win 52% of votes nationally, algorithmic plans producing 56.5% Democratic seats show modest over-representation. Achieving strict proportionality may require:

- Relaxing compactness requirements to allow strategic district shapes
- Abandoning single-member districts in favor of multi-member districts with proportional voting
- Accepting geographic constraints as fundamental limitations

The tension between compact districts and proportional outcomes is not a failure of algorithmic methods—it reflects the geographic structure of the American electorate Chen and Rodden (2013).

5.5 Limitations

1. **Precinct aggregation**: Efficiency gaps computed from precinct results allocated to districts. Block-level analysis would be more precise but computationally intensive.
2. **Election selection**: We use presidential and midterm elections. Off-year or local elections might show different patterns.
3. **Single algorithm**: Results based on recursive bisection. Other algorithmic approaches (ReCom, shortest splitline) might produce different efficiency gaps.
4. **Competitive states focus**: Detailed analysis limited to 15 competitive states. Non-competitive states (e.g., Wyoming, Vermont) omitted as efficiency gap less meaningful in landslide contexts.

6 Conclusion

This paper provides the first national-scale efficiency gap analysis of algorithmic redistricting, establishing empirical benchmarks for partisan fairness evaluation. Our findings demonstrate that neutral algorithmic methods substantially reduce partisan bias—62% lower than enacted plans—while maintaining constitutional requirements and geographic compactness. The persistent -3.2% efficiency gap in algorithmic plans reflects unavoidable Democratic urban concentration and establishes a lower bound on partisan asymmetry achievable through neutral redistricting.

These results have important implications for three constituencies. For **reformers**, algorithmic redistricting offers meaningful improvements in partisan fairness without requiring constitutional amendments or abandoning traditional redistricting principles. The 62% bias reduction represents substantial progress toward neutral governance even if perfect proportionality remains elusive. For **courts**, efficiency gap benchmarks provide quantitative standards for evaluating partisan gerrymandering claims under state constitutions. States showing enacted efficiency gaps substantially exceeding algorithmic baselines (e.g., $+7\%$ vs. -3%) exhibit evidence of human manipulation beyond geographic necessity. For **scholars**, our analysis clarifies the limits of algorithmic objectivity: neutral algorithms reduce but cannot eliminate partisan effects, and the residual -3.2% bias reflects fundamental geographic constraints rather than algorithmic failures.

Future research should extend this analysis in three directions. First, comparing efficiency gaps across different algorithmic approaches (ReCom, shortest splitline, simulated annealing) would test whether our findings are method-specific or represent general properties of neutral redistricting. Second, analyzing efficiency gaps at finer geographic resolutions (census blocks rather than tracts) would increase precision and potentially reveal whether algorithmic bias can be further reduced through more granular optimization.

Third, exploring multi-member district alternatives would quantify the partisan fairness gains achievable by moving beyond single-member districts entirely.

The gerrymandering crisis identified by Chief Justice Roberts in *Rucho* remains acute: partisan manipulation of district boundaries undermines democratic legitimacy and entrenches political power. While *Rucho* removed federal judicial oversight, state courts and legislatures retain authority to address partisan gerrymandering. Our analysis demonstrates that algorithmic redistricting provides a practical, implementable solution that substantially reduces partisan bias while respecting constitutional constraints. The question is no longer whether neutral alternatives exist—they do, and this paper quantifies their partisan fairness advantages. The question is whether political institutions will adopt them.

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